

1. DEFINITION

DEFINITION
Emulsifier mixture of: PEG-6 stearate (type I) NF and PEG-32 stearate (type I) NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	PEG-6 Stearate (and) PEG-32 Stearate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	8LQC57C6B0 33GX5WQC0M
ADDITIONAL INFORMATION ON THE DENOMINATION	Preferred Substance Name: PEG-32 stearate UNII: 33GX5WQC0M Other names: Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxyoctadecanoate, Polyoxyl 32 Palmitostearate, Polyoxyl Stearate
	Preferred Substance Name: PEG-6 Stearate UNII: 8LQC57C6B0 Other names: Poly(oxy-1,2-ethanediyl), α -hydro- ω -hydroxyoctadecanoate, Polyethylene glycol 300 monostearate, Polyoxyethylene (6) monostearate, Polyoxyl 6 Palmitostearate, Polyoxyl Stearate
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Simple physical mixing of the 2 components: PEG-6 palmitostearate and PEG-32 palmitostearate



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COMPOSITION	
Simple physical mixture of 2 components: PEG-6 palmitostearate and PEG-32 palmitostearate	
Full Labeling (Key letters system)	
$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
PEG-6 Stearate (and) PEG-32 Stearate	$50.0\% \leq A_2 < 75.0\%$ $25.0\% \leq B < 50.0\%$
Other constituents (i.e. incidental ingredients)	
None	

4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	6074
Canada MF (Type III – Excipient)	2011-249 (inactive)
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS		
Reference	Component	Conforms to monograph
European Pharmacopeia Current edition	PEG-6 Stearate	/
	PEG-32 Stearate	/
USP/ NF current edition	PEG-6 Stearate	Polyoxyl Stearate (type I)
	PEG-32 Stearate	Polyoxyl Stearate (type I)
Japanese Pharmaceutical Excipients JPE Current edition	PEG-6 Stearate	/
	PEG-32 Stearate	/



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6. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	PEG-6 stearate Polyethylene glycol 300 monostearate	Available for use as an Excipient Ingredient in: Export Only	none
Approved	PEG-32 stearate Polyethylene glycol 1540 monostearate	Available for use as an Excipient Ingredient in: Export Only	None
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	PEG-6 Stearate	<u>Role Non-medicinal:</u> Purposes: Emulsifying Agent Route Of Administration: Topical Detail: Topical use only. Reference: CIR
Approved chemical name	PEG-32 Stearate	<u>Role Non-medicinal:</u> Purposes: Emulsifying Agent Route Of Administration: Topical Detail: Topical use only, up to 10%
Approved chemical name	Polyoxyethylene stearates	<u>Role non-medicinal:</u> Purposes: Emulsifying Agent , Solubilizing Agent , Wetting Agent Detail: Acceptable daily intake up to 25 mg/kg body weight daily as total for polyoxyethylene derivatives
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

7. SAFETY ASSESSMENT



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SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (JORF)	Dermal	Rabbit	TEFOSE® 1500 (PEG-6 + PEG-32 stearate) No dilution	Non Irritant Cutaneous irritation (24h): 0.24	(Gattefossé data, GAT-7377, 1973)
Acute irritation (OECD 404)	Dermal	Rabbit	TEFOSE® 1500 (PEG-6 + PEG-32 stearate) 20% in water	Moderately irritant	(Gattefossé data, GAT-99468, 1999)
Acute irritation (JORF)	Ocular	Rabbit	TEFOSE® 1500 (PEG-6 + PEG-32 stearate) 20% in water	Non irritant Ocular irritation (24h): 3.67	(Gattefossé data, GAT-7377, 1973)
Acute irritation (OECD 405)	Ocular	Rabbit	TEFOSE® 1500 (PEG-6 + PEG-32 stearate) No dilution	Moderately irritant	(Gattefossé data, GAT-99495, 1999)
Acute toxicity (OECD 401)	Oral	Rat	TEFOSE® 1500 (PEG-6 + PEG-32 stearate) No dilution Single dose of 2006 mg/kg	LD50 _(oral) > 2006 mg/kg	(Gattefossé data, GAT-99469, 1999)
Suchronic 3-month toxicity study	Dermal	Rabbit	TEFOSE® 1500 (PEG-6 + PEG-32 stearate) 20% in water	NOAEL _(dermal) > 2000 mg/kg Very Well Tolerated	(Gattefossé data, GAT-7377, 1973)

OECD : Organisation for Economic Co-operation and development - JORF: Journal officiel République Française (Official Journal French Republic)

Summary of files available upon request, please contact your Gattefossé local representative.
Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview"
for a complete review on regulatory, tox and safety aspects.

CIR COMPENDIUM (Cosmetic Ingredient Review)

(<http://www.cir-safety.org>)

Denomination	Reference	Maximum "as used" concentration	Conclusions
PEG Stearates	JACT 2(7): 17-60, 1983	Up to 5% (PEG-6 stearate) Up to 10% (PEG-32 Stearate)	On the basis of the available information presented in this report, the panel concludes that PEG-2, -6, -8, - 12, -20, -32, -40, -50, -100 and -150 Stearates are safe as cosmetic ingredients in the present practices of concentration and use.



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8. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	CAS #	EC #	Chemical name
PEG-6 Stearate	9004-99-3	Not applicable	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-hydroxy-
PEG-32 Stearate	9004-99-3	Not applicable	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-hydroxy-

EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACH submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.

NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
PEG-6 Stearate	Listed	Listed	Listed	Listed	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-hydroxy-: AIIC, DSL, ENCS, IECSC, INSQ, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI, AREC, ECL
PEG-32 Stearate	Listed	Listed	Listed	Listed	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-hydroxy-: AIIC, DSL, ENCS, IECSC, INSQ, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI, AREC, ECL

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
PEG-6 Stearate (and) PEG-32 Stearate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.



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9. HISTORY OF USE

9.1. Pharmaceutical applications

9.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
POLYOXYL 6 AND POLYOXYL 32 PALMITOSTEARATE (UNII: NA)	TOPICAL	Cream	20.0 % w/w	
	TOPICAL	Lotion	5 % w/w	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

9.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
POLYOXYL STEARATE (UNII: NA)	AURICULAR (OTIC)	SUSPENSION	0.01%w/w	
	ORAL	TABLET, COATED	2mg	
	TOPICAL	AEROSOL, FOAM	4.36%w/w	
	TOPICAL	CREAM		70mg
	TOPICAL	EMULSION	NA	
	TOPICAL	LOTION	2%w/w	
	VAGINAL	CREAM	0.25% w/w	
PEG-8 STEARATE (UNII: 2P9L47VI5E)	ORAL	CONCENTRATE	25mg/5ml	
	ORAL	SUSPENSION		60 mg
	TOPICAL	CREAM		229 mg
	TOPICAL	SPONGE	NA	



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FDA INACTIVE INGREDIENT GUIDE

(<http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm>)

Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
POLYOXYL-40 STEARATE (UNII: 13A4J4NH9I)	AURICULAR (OTIC)	SUSPENSION	1%w/v	
	OPHTHALMIC	OINTMENT	NA	
	OPHTHALMIC	SOLUTION	7%w/w	
	OPHTHALMIC	SUSPENSION	0.5%w/v	
	ORAL	CAPSULE	7mg	
	ORAL	GRANULE	NA	
	ORAL	TABLET	8.48mg	
	ORAL	TABLET, FILM COATED	2mg	
	TOPICAL	AEROSOL, FOAM	2.83%w/w	
	TOPICAL	CREAM	8.8%w/w	
	TOPICAL	LOTION	3%w/w	
	TOPICAL	OINTMENT	NA	
	TOPICAL	SPONGE	NA	

Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.

9.1.3. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS

(Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)

Denomination	Functional Category	Regulatory status	Safety
Polyoxyethylene stearates	Emulsifying agent; Solubilizing agent; Wetting agent.	Included in the FDA Inactive Ingredients Database (dental solutions; IV injections; ophthalmic preparations; oral capsules and tablets; otic suspensions; topical creams, emulsions, lotions, ointments, and solutions; and vaginal preparations). Included in nonparenteral medicines licensed in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.	Although polyoxyethylene stearates are primarily used as emulsifying agents in topical pharmaceutical formulations, certain materials, particularly polyoxyl 40 stearate, have also been used in intravenous injections and oral preparations. Polyoxyethylene stearates have been tested extensively for toxicity in animals and are widely used in pharmaceutical formulations and cosmetics. They are generally regarded as essentially nontoxic and non-irritant materials. Polyoxyl 8 stearate LD50 (hamster, oral): 27 g/kg LD50 (rat, oral): 64 g/kg Polyoxyl 20 stearate LD50 (mouse, IP): 0.2 g/kg LD50 (mouse, IV): 0.87 g/kg



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COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	Category (concentration in %)	regulatory status
Polyoxyl Stearates	Emulsifying agent	1, 4, 10, 12, 16, 23–25
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portugese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., <i>Pharmaceutical Technology, Drug Delivery</i>, 2001, p14–24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		

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